All integer-power counterexamples for the Jain–Karlin–Schoenberg kernel

Automated mathematics run `fa_banach_001`, lane 10

17 August 2026

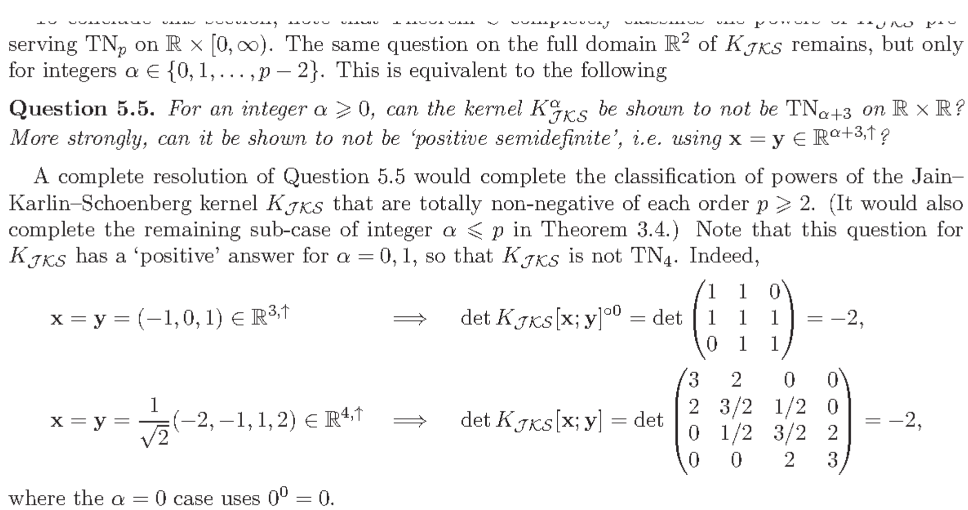
Status: candidate full solution, likely valid; novelty not certified. For every integer $m \geq 0$ an explicit increasing $(m+3)$-tuple gives a symmetric Jain–Karlin–Schoenberg Gram matrix whose entrywise mth power has negative determinant. This answers both parts of Question 5.5 in arXiv:2008.05121 and, together with Theorem C there, closes the full-plane TN_p power classification.

1 Source question

Khare's paper [1] studies the symmetric kernel

$$K_{\mathrm{JKS}}(x,y) := \max(1+xy, 0), \qquad x, y \in \mathbb{R}.$$

Question 5.5, in Section 5 on printed page 19 of arXiv:2008.05121v2, asks for every integer $\alpha \geq 0$ whether $K_{\mathrm{JKS}}^{\circ\alpha}$ can be shown not to be $\mathrm{TN}_{\alpha+3}$ on $\mathbb{R} \times \mathbb{R}$. Its stronger form asks for a witness with the same increasing tuple in both arguments, equivalently a symmetric matrix that is not positive semidefinite. As in the source, $0^0 = 0$.


serving TN_p on $\mathbb{R} \times [0,\infty)$. The same question on the full domain $\mathbb{R}^2$ of $K_{\mathcal{JKS}}$ remains, but only for integers $\alpha \in \{0,1,\dots,p-2\}$. This is equivalent to the following

Question 5.5. *For an integer $\alpha \geqslant 0$, can the kernel $K_{\mathcal{JKS}}^{\alpha}$ be shown to not be $\mathrm{TN}_{\alpha+3}$ on $\mathbb{R} \times \mathbb{R}$? More strongly, can it be shown to not be 'positive semidefinite', i.e. using $\mathbf{x} = \mathbf{y} \in \mathbb{R}^{\alpha+3,\uparrow}$?*

A complete resolution of Question 5.5 would complete the classification of powers of the Jain–Karlin–Schoenberg kernel $K_{\mathcal{JKS}}$ that are totally non-negative of each order $p \geqslant 2$. (It would also complete the remaining sub-case of integer $\alpha \leqslant p$ in Theorem 3.4.) Note that this question for $K_{\mathcal{JKS}}$ has a 'positive' answer for $\alpha = 0, 1$, so that $K_{\mathcal{JKS}}$ is not TN_4. Indeed,

$$\mathbf{x} = \mathbf{y} = (-1,0,1) \in \mathbb{R}^{3,\uparrow} \qquad \implies \qquad \det K_{\mathcal{JKS}}[\mathbf{x};\mathbf{y}]^{\circ 0} = \det \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} = -2,$$

$$\mathbf{x} = \mathbf{y} = \frac{1}{\sqrt{2}}(-2,-1,1,2) \in \mathbb{R}^{4,\uparrow} \qquad \implies \qquad \det K_{\mathcal{JKS}}[\mathbf{x};\mathbf{y}] = \det \begin{pmatrix} 3 & 2 & 0 & 0 \\ 2 & 3/2 & 1/2 & 0 \\ 0 & 1/2 & 3/2 & 2 \\ 0 & 0 & 2 & 3 \end{pmatrix} = -2,$$

where the $\alpha = 0$ case uses $0^0 = 0$.

Figure 1: Question 5.5 and the two low-degree examples in the source paper, printed page 19.

2 Proof intuition

The substitution $x = \tan u$ converts the kernel, up to harmless positive diagonal factors, into the positive part of the cosine difference kernel. Choose $m+3$ equally spaced angles whose total span is just over $\pi/2$, but whose every shorter pairwise span is below $\pi/2$. Truncation then deletes exactly the two opposite corner entries of the untruncated cosine-power matrix.

The untruncated matrix has a finite Fourier Gram representation with only $m+1$ frequencies, hence rank $m+1 = (m+3)-2$. Its middle $(m+1)$-square block is nevertheless positive definite, by a square Vandermonde determinant. Expanding after deleting the two corners, the constant and linear terms vanish by the rank deficiency; the unique quadratic term is the negative of the middle determinant. This gives an exact negative determinant in every degree, including $m = 0$.

3 The construction and determinant identity

Theorem 1 (Symmetric witness in every integer degree)**.** *Let $m \in \mathbb{Z}_{\geq 0}$ and put $n = m+3$. Choose any*

$$\frac{\pi}{2(n-1)} < \delta < \frac{\pi}{2(n-2)}, \qquad u_j = \left(j - \frac{n+1}{2}\right)\delta, \qquad x_j = \tan u_j \quad (1 \leq j \leq n).$$

Then $x_1 < \cdots < x_n$, and the symmetric matrix

$$M = (K_{\mathrm{JKS}}(x_j, x_k)^m)_{j,k=1}^n$$

has negative determinant. Consequently it is not positive semidefinite, and $K_{\mathrm{JKS}}^{\circ m}$ is not TN_{m+3} *on* $\mathbb{R} \times \mathbb{R}$*.*

Proof. Write $L = (n-1)\delta$. The choice of δ gives

$$\frac{\pi}{2} < L < \pi.$$

Indeed, the second inequality is immediate when $n = 3$, and for $n > 3$ follows from $(n-1)/(2(n-2)) < 1$. Hence all u_j lie in $(-\pi/2, \pi/2)$, so the $x_j = \tan u_j$ are well-defined and strictly increasing.

For $u, v \in (-\pi/2, \pi/2)$,

$$K_{\mathrm{JKS}}(\tan u, \tan v) = \sec u \, \sec v \, [\cos(u-v)]_+. \tag{1}$$

Define

$$A_{jk} := [\cos(u_j - u_k)]_+^m, \qquad C_{jk} := \cos^m(u_j - u_k).$$

For $m = 0$, the first definition uses the source convention $0^0 = 0$, while the second has the usual value $a^0 = 1$ because all occurring cosines in C are nonzero. Every pair other than $\{1, n\}$ has separation at most $(n-2)\delta < \pi/2$, whereas $|u_n - u_1| = L > \pi/2$. Thus truncation changes exactly the two opposite corners. With

$$c := \cos^m L$$

(so $c = 1$ when $m = 0$), we have

$$A = C - c(E_{1n} + E_{n1}). \tag{2}$$

The elementary Fourier expansion

$$\cos^m t = 2^{-m} \sum_{r=0}^{m} \binom{m}{r} e^{i(m-2r)t} \tag{3}$$

expresses C as VDV^*, where

$$V_{jr} = e^{i(m-2r)u_j} \quad (0 \le r \le m), \qquad D_{rr} = 2^{-m}\binom{m}{r} > 0.$$

Therefore C is positive semidefinite and $\operatorname{rank} C \le m+1 = n-2$.

Let

$$B = C[\{2,\dots,n-1\},\{2,\dots,n-1\}].$$

This is the Gram matrix from the middle $m+1$ rows of V. That square submatrix of V is nonsingular: after extracting e^{imu_j} from row j, it is the Vandermonde matrix

$$\left(z_j^r\right)_{j=2,\dots,n-1;\ r=0,\dots,m}, \qquad z_j = e^{-2iu_j}.$$

The nodes are distinct, since $0 < |u_j - u_k| < \pi$ for $j \ne k$. Hence B is positive definite and

$$\det B > 0. \tag{4}$$

It remains to expand (2). Regard the determinant as a polynomial in the two changes at positions $(1,n)$ and $(n,1)$. Its constant term is $\det C = 0$. Each linear coefficient is an $(n-1)$-minor of C and hence vanishes because $\operatorname{rank} C \le n-2$. The coefficient of the product of the two changes is $-\det B$: a contributing permutation exchanges the two endpoint columns, producing one minus sign, and its remaining part is the determinant of the middle block. Since both changes equal $-c$,

$$\det A = -c^2 \det B < 0. \tag{5}$$

Here $c \ne 0$ because $L \in (\pi/2, \pi)$; this also covers $m = 0$, when $B = [1]$ and (5) reads $\det A = -1$.

Finally, (1) yields $M = D_m A D_m$ with $D_m = \operatorname{diag}(\sec^m u_1, \dots, \sec^m u_n)$. Thus

$$\det M = \left(\prod_{j=1}^{n} \sec(u_j)^{2m}\right) \det A < 0.$$

A positive-semidefinite matrix has nonnegative determinant, so M is not positive semidefinite. Its negative full minor also violates $\mathrm{TN}_n = \mathrm{TN}_{m+3}$. □

4 The completed full-plane classification

Corollary 1. *For every integer $p \ge 2$ and every real $\alpha \ge 0$,*

$$K_{\mathrm{JKS}}^{\circ\alpha} \text{ is } \mathrm{TN}_p \text{ on } \mathbb{R}\times\mathbb{R} \quad \Longleftrightarrow \quad \alpha \ge p-2.$$

Proof. Theorem C(1) of [1] proves the forward construction for every $\alpha \ge p-2$. Theorem C(2) there proves that a TN_p power below $p-2$ would have to be an integer. If such an integer is m, then $m \le p-3$, and Theorem 1 gives an $(m+3)$-minor with negative determinant. Since $m+3 \le p$, this contradicts TN_p. □

5 Verification

The exact argument was audited at the following points.

- The interval for δ is nonempty and places exactly one unordered endpoint pair beyond the cosine cutoff.
- Formula (3) gives at most $m + 1 = n - 2$ Gram vectors with strictly positive weights.
- The middle evaluation matrix is a square Vandermonde matrix on distinct unit-circle nodes, so no hidden singularity remains.
- A direct determinant expansion confirms both the sign and the middle minor in (5).
- The convention at $m = 0$ was checked separately and is also contained in the same formula.

The reusable script `code/verify_jks_family.py` evaluates the matrices at 180-digit precision for $m = 0, \ldots, 16$. It checks $\det B > 0$, $\det A < 0$, and $\det A = -\cos^{2m}(L) \det B$ to relative error below 10^{-70}. All 17 cases pass. This computation is a regression check, not part of the proof.

6 Novelty search, limitations, and review recommendation

The run's registry, solution, attempt, and proof-gap indexes were searched for the arXiv id, title, kernel name, and core formula, with no duplicate found. A bounded external search through 17 August 2026 used the exact Question 5.5 label, "Jain–Karlin–Schoenberg kernel", `max(1+xy,0)`, "integer powers", "alpha+3", and "positive semidefinite". The current author-hosted PDF still states the question. Close later arXiv papers checked included 2103.12550, 2110.08206, and 2411.03391; none was found to state this construction or resolve Question 5.5. Novelty is therefore plausible but is not certified by this bounded search.

There is no conditional dependency in the proof. The main human-review focus should be the two-entry determinant coefficient in (5) and the inference in Corollary 1 from an $(m + 3)$-minor to failure of TN_p. Subject to that check and a specialist literature search, promote as a full solution.

References

[1] A. Khare, *Multiply positive functions, critical exponent phenomena, and the Jain–Karlin–Schoenberg kernel*, arXiv:2008.05121v2, Question 5.5 and Theorem C (2020–2021).